

ECE 269: Linear Algebra and Applications
Winter 2026
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Homework # 4

1. **Problem 1: Moore–Penrose pseudoinverse.** A pseudoinverse of $A \in \mathbb{R}^{m \times n}$ is defined as a matrix $A^+ \in \mathbb{R}^{n \times m}$ that satisfies

$$AA^+A = A, \quad A^+AA^+ = A^+$$

and AA^+ and A^+A are symmetric.

- (a) Show that A^+ is unique.
 - (b) Show that $(A^T A)^{-1} A^T$ is the pseudoinverse and a left inverse of a full-rank tall matrix A .
 - (c) Show that $A^T (A A^T)^{-1}$ is the pseudoinverse and a right inverse of a full-rank fat matrix A .
 - (d) Show that A^{-1} is the pseudoinverse of a full-rank square matrix A .
 - (e) Show that A is the pseudoinverse of itself for a projection matrix A .
 - (f) Show that $(A^T)^+ = (A^+)^T$.
 - (g) Show that $(A A^T)^+ = (A^+)^T A^+$ and $(A^T A)^+ = A^+ (A^+)^T$.
 - (h) Show that $\mathcal{R}(A^+) = \mathcal{R}(A^T)$ and $\mathcal{N}(A^+) = \mathcal{N}(A^T)$.
 - (i) Show that $P = A A^+$ and $Q = A^+ A$ are projection matrices.
 - (j) Show that $y = P x$ and $z = Q x$ are the projections of x onto $\mathcal{R}(A)$ and $\mathcal{R}(A^T)$, respectively, where P and Q are defined as previously.
 - (k) Show that $x^* = A^+ b$ is a least-squares solution to the linear equation $Ax = b$, i.e., $\|Ax^* - b\| \leq \|Ax - b\|$ for every other x .
 - (l) Show that $x^* = A^+ b$ is the least-norm solution to the linear equation $Ax = b$, i.e., $\|x^*\| \leq \|x\|$ for every other solution x , provided that a solution exists.
2. **Problem 2: Eigenvalues.** Suppose that $A \in \mathbb{C}^{n \times n}$ has $\lambda_1, \lambda_2, \dots, \lambda_n$ as its eigenvalues repeated according to their algebraic multiplicities.
- (a) Show that $\det(A) = \lambda_1 \cdot \lambda_2 \cdots \lambda_n$.
 - (b) Show that $\text{Trace}(A) = \lambda_1 + \lambda_2 + \cdots + \lambda_n$.
 - (c) Show that the eigenvalues of A^k are $\lambda_1^k, \lambda_2^k, \dots, \lambda_n^k$ for $k = 1, 2, \dots$.
 - (d) Show that A is invertible if and only if it does not have a zero eigenvalue.

(e) Suppose that A is invertible. Show that the eigenvalues of A^{-1} are $\lambda_1^{-1}, \lambda_2^{-1}, \dots, \lambda_n^{-1}$.

3. **Problem 3: Nilpotent Matrices.** A matrix $A \in \mathbb{C}^{n \times n}$ is said to be nilpotent if $A^k = 0$ for some integer k .

(a) Show that all eigenvalues of a nilpotent matrix must be 0.

(b) Show that the smallest k for which $A^k = 0$, satisfies $k \leq n$.

(c) Suppose that the smallest k for which $A^k = 0$, is $k = n$. Further suppose that $A^{n-1}x \neq 0$. Then show that $\{x, Ax, A^2x, \dots, A^{n-1}x\}$ form a basis of $\mathbb{C}^n$.

4. **Problem 4: Spectral Norm.** Define spectral norm $\|A\|$ of a matrix A as

$$\|A\| := \max_{x \neq 0} \frac{\|Ax\|_2}{\|x\|_2}$$

(a) Show that $\|A^H A\| = \|A\|^2$.

(b) Show that the spectral norm is *unitarily invariant*, namely, $\|UAV\| = \|A\|$ for any unitary matrices U and V .

(c) Show that

$$\left\| \begin{bmatrix} A & 0 \\ 0 & B \end{bmatrix} \right\| = \max(\|A\|, \|B\|).$$

5. **Problem 5: Properties of PSD matrices.** Let $A = A^T \in \mathbb{R}^{n \times n}$ and $B = B^T \in \mathbb{R}^{n \times n}$. The relation $A \succeq B$ means that $A - B$ is PSD. Prove the following statements.

(a) If $A \succeq 0$ and $B \succeq 0$, then $\text{Trace}(AB) \geq 0$.

(b) If $A \succeq 0$, then $A + B \succeq B$.

(c) If $A \succeq B$, then $-B \succeq -A$.

(d) If $A \succeq I$, then $I \succeq A^{-1}$.

(e) If $A \succeq B \succ 0$, then $B^{-1} \succeq A^{-1} \succ 0$.